

Biochar: Kiln, Quality, Storage

Carpathian OER Commons — Land, Forest, Food Module

Michel Garand | 2026-09-29 | Version v0.1 | CC BY-SA 4.0

Maturity: L1 — designed and costed, untested. Every cost line is an estimate or a listed price, labelled as such in Section 6; one is still costed at the working paper's 25 people. No kiln has been built and no char made. Read every figure below at this level and no higher.

No burn is made before the written answers on the kiln and on the wood (Section 10), and never in a fire-danger period, a dry spell, or against a standing order of the authorities.

Governed by Section 2 of the compendium. No child appears in this module in any form. The site is not named; site plans are schematic; photographs show structures and systems only. Design loads are site capacity, used for sizing only.

Converted from the sanitation working paper, EN v0.2, Sections 2 and 8.2. The paper sized char for the toilets alone, at 25 people; this module sizes it for both streams at 34, and replaces the paper's kiln line with a flame-curtain cone welded locally to a published design.

1. Purpose

Char for both nutrient streams, made on the site from its own woody clearings: a scoop at every use of the bucket toilets, and 10 to 30 percent of every garden compost pile. One kiln, two exits, and nothing burned that the clearing should have left.

Six rules. One kiln, two streams: raw crushed char goes to the seat and to the garden pile, and each stream charges it; no raw char goes onto a bed. Dry wood only: feedstock dried below about 20 percent moisture, because wet wood makes methane. Burn what is cleared, never what is left: standing and fallen deadwood stays as habitat, as the programme proposal holds, and the kiln takes only what the clearing takes. The fire keeps its distances and its days: outside every distance in Section 4, never unattended, never in a fire-danger period or a dry spell. Water for the quench from the pond or the rain, never the spring. The work is paid; toloka is not the name for work the kiln depends on.

Connections. Depends on: the clearing done by the planting module and the regeneration strand, which sets the feedstock; the land's category and tenure, from the land register, which decides whether a felling permit is needed; the written answers on the kiln; the market garden's pond, or the rainwater module, for the quench; the Phase 1 walk, with the fire distances measured. Feeds: the sanitation toilets row, with char at the seat, which the terra preta module receives in every vessel; the compost module, with char for the garden pile.

Patterns. Grows from: not yet named. Which trees this valley does not take, and what it holds about wood from the forest, are for the elders before the first burn; the literature records forked trees, trees with "candles", and windthrow left unused for building [for the elders]. Feeds: not yet named. Resonates with: 104 Site Repair — the kiln on bare, poorer ground outside every fire distance; 172 Garden Growing Wild — the kiln burns what the clearing takes and never what it leaves; 120 Paths and Goals — the kiln on the line cleared material already travels. In tension with: 181 The Fire — a real fire, kept

at the edge and never at the centre; 7 The Countryside — wood taken only where the land's tenure and the valley's custom allow; 64 Pools and Streams — the quench drawn from the pond or the rain, never from the spring.

2. Context of validity

Condition	Range this design holds for
Climate	[set] — burns in calm, damp weather only
Altitude	[set]
Feedstock	brush and branches under about 15 cm across
Moisture	below about 20 percent, air-dried
Need	the toilets at capacity; the garden's piles
Group size	capacity: 34 people, the example below
Ground	bare, level, outside every fire distance
Quench water	the pond or rainwater, about a kiln's volume

3. Sizing method

What you must measure yourself before using this module:

- **Your char need** — the toilets' char at the seat, from the terra preta module, and the garden's char, from the compost module.
- **Your feedstock** — the woody material the clearing actually yields in a year, stacked and measured, never estimated from the forest.
- **Your moisture** — the feedstock's, before a burn, by a moisture meter or by weighing a sample before and after drying.
- **Your yield** — char out, in litres, against feedstock in, for the first burns.
- **Your distances** — as in Section 4, measured with a tape.

Our example: char_at_seat, 0.5 L a day; bucket_share, two-thirds; capacity, 34 people; use_days_per_year, 120; char_density, 186 kg per m³; char_yield, 15 to 22 percent of dry mass. Values from the site parameters file, held in the workshop.

3.1 The need

Stream	Calculation	Our example
The toilets	$34 \times \text{two-thirds} \times 0.5 \text{ L} \times 120$	about 1.4 m ³
The garden	10–30 % of 2.4–14.4 m ³ raw	about 0.2–4.3 m ³
A year	sum	about 1.6–5.7 m³
By mass	$\times 186 \text{ kg per m}^3$	about 300–1,060 kg

The garden's raw piles, 2.4 to 14.4 m³ a year, are the compost module's kitchen and garden streams before they shrink; its char is 10 to 30 percent by volume, the Ithaka Institute's range (Schmidt, 55 uses of biochar). The paper's 1.5 m³ a year was for every use at 25 people; the toilets' share at 34 with two-thirds of uses in buckets is about 1.4 m³.

3.2 The wood and the burns

Quantity	Calculation	Our example
Dry wood, a year	$300\text{--}1,060 \text{ kg} \div 15\text{--}22 \%$	about 1.4–7.0 t
Char a burn	the kiln's volume, roughly	up to about 1 m ³
Burns, a year	$1.6\text{--}5.7 \text{ m}^3 \div \text{about } 1 \text{ m}^3$	about 2–6
Loose brush a burn	ten times the kiln's volume	about 8 m ³
Hours a burn	as the feedstock allows	2 to 8

Yield. Flame-curtain kilns gave 22 ± 5 percent of dry mass, 20 ± 4 when quenched with water (Cornelissen and colleagues, PLOS ONE, 2016); Schmidt and Taylor give 15 to 20 percent, and two to eight hours to make roughly 1 m³ of char in their larger cone (Kon-Tiki, the Biochar Journal, 2014). A flame-cap kiln uses about ten times its volume in feedstock (South Umpqua Rural Community Partnership, 2018). Char is taken at 186 kg per m³, measured from a flame-cap kiln in the field (Wilson and colleagues, 2024); biochars range from 80 to 320 kg per m³.

The feedstock decides. At the upper end the need is about 7 t of dry wood a year. The programme proposal leaves standing and fallen deadwood as habitat and clears only what chokes the young growth; if the clearing yields less than the need, the garden's char is cut first, and the toilets' char is kept.

3.3 The kiln, as designed

Element	Specification
Form	a cone, open at the top, flame-curtain
Rim	1.5 m across
Depth	0.9 m
Wall	63° from the horizontal
Base	about 0.58 m across, derived
Volume	about 0.82 m ³ , derived
Steel	3 mm, hot-rolled; about 3.6 m ² , 84 kg, derived
Quench	by bucket from the water store
Alternative	the same cone dug in mineral soil

The rim, depth, and angle are those of Schmidt and Taylor's first steel cone of about 750 L; the base, volume, area, and mass are derived from them here, not quoted. The published steel is 2 mm for small cones and 3 mm for larger ones, a figure to be confirmed against the source. The design is published by its authors as open source.

4. Bill of materials

Quantities for the example.

Item	Quantity	Alternatives in the design
Steel sheet, 3 mm, 1.25 × 2.5 m	2	the soil pit
Fabrication, cut, rolled, welded	about 84 kg	the soil pit
Garden sieve, 10 mm insert	1	welded mesh
Respirators, FFP2 with valve	20 a year, [assumed]	none named
Goggles, closed	2	none named
Char and cover store	1, dry, covered	none named

Item	Quantity	Alternatives in the design
Quench water store, 1,000 L	1	the pond, by pump
Woven bags for crushing	[set], from the site	a hammer mill

Crushing. Wet, in closed woven bags, under a roller or a sledgehammer, to less than 10 mm, then through the sieve (IBI, Guidelines on Practical Aspects of Biochar Application, 2010; Permachar Kontiki manual, 2023). Dry char dust can form explosive mixtures with air in closed spaces, and tightly packed char can heat itself (IBI, 2010): the store keeps char damp, near the 30 percent water the European Biochar Certificate recommends for storage, and loosely heaped. A grain hammer mill, listed at 2,799 UAH, is the alternative for larger batches; wet char clogs it.

The siting, as designed. Every distance below is the larger of what the instruments in Section 10 ask, and every one is measured on the ground before the first burn; none has been measured yet.

From the kiln to	At least
Any building	30 m
Coniferous forest	50 m
Deciduous forest	25 m
The spring, and never upslope of it	50 m
The stream and the pond, on a slope	50 m
Any place where children are	30 m
A cleared, non-combustible strip round it	2.5 m wide

The layout, as designed. The kiln on bare, poorer ground, on the line cleared material already travels; the feedstock stacked to dry under cover, upwind and outside the cleared strip; the water store beside the kiln; the char store between the kiln and its two uses. Drawn as relationships only; the walk sets the rest.

5. Build

Step	Trade	Time	If done out of order
Written answers on kiln, wood	household	[from L2]	no burn

Step	Trade	Time	If done out of order
Walk; fire distances measured	household	[from L2]	kiln too near
Cone cut, rolled, welded	welder	[from L2]	[from L2]
Kiln ground cleared, strip	household	[from L2]	fire spreads
Water store set and filled	household	[from L2]	no quench
Feedstock stacked to dry	responsible adult	a season	methane
First burn, yield measured	responsible adult	[from L2]	[from L2]

A burn, as published. An open stack of dry wood lit at the top; the next layer added when the top of the last begins to coat with ash; the flame kept strong so that it burns its own smoke; the char quenched with water when the last layer has charred, until it floats (Schmidt and Taylor, 2014; Permachar, 2023). Smoke means too much added at once (South Umpqua, 2018).

6. Cost

Estimates, not quotations: each line is either a price listed by a Ukrainian retailer or service, seen on 2026-09-29, or the sanitation paper's own estimate for autumn 2026, marked "paper, at 25". Place: not a quotation; western Ukrainian listings where found, national otherwise; the steel at a national supplier's price of 28.09.2026. Exchange rate: exchange_rate, 52 UAH to the euro, the National Bank's mid-September 2026 rate, rounded.

6.1 The kiln

Item	UAH	EUR	Basis
Steel, 2 sheets 3 mm at 3,710.50	7,421	143	estimated, listing
Fabrication, 84 kg at 34,880 a t	2,934	56	listed average rate
Subtotal	10,355	199	

6.2 Crushing, safety, and storage

Item	UAH	EUR	Basis
Garden sieve, 10 mm insert	600	12	estimated, listing

Item	UAH	EUR	Basis
Respirators, 20 at 41	820	16	listing; qty assumed
Goggles, 2 at 206	412	8	estimated, listing
Char and cover store	15,000	288	paper, at 25
Quench water store, 1,000 L	4,000	77	listing, out of stock
Subtotal	20,832	401	

6.3 Summary

Section	UAH	EUR
The kiln	10,355	199
Crushing, safety, and storage	20,832	401
Construction subtotal	31,187	600
Design check, 8 percent	2,500	48
Contingency, 20 percent	6,200	119
Total external cost	39,887	767

Against the paper's 53,000 UAH for the kiln, crusher, and store, the kiln is welded locally from its published design instead of bought or built to an unnamed one. For comparison only, a small flame-curtain kiln of about 80 L of char a cycle is listed in Germany at 1,100 EUR with VAT (Prodana, seen 2026-09-29); no priced small kiln was found in Ukraine. The char and cover store is still the paper's estimate at 25 people, now holding a year's char for both streams and the toilets' cover, and is marked for a new estimate. The percentages are the paper's own.

Contribution, not in the total. None counted. The feedstock comes from the site's own clearing; the work of gathering, stacking, and burning is paid.

Running cost. The responsible adult's hours — two to six burns of two to eight hours, with gathering, stacking, crushing, and sieving — [set], paid at no less than the legal minimum wage, 8,647 UAH a month in 2026, for the hours worked. Respirators, 820 UAH a year. The kiln itself is steel: its life is [from L2].

7. Operation and maintenance

Responsible: one named adult, paid, with a deputy — [from L2]. Not a rota. The kiln is never left unattended.

- **Before a burn:** the day checked — no fire-danger period, no dry spell of ten days or more, calm or damp weather, no standing order against fire; the water store full; the cleared strip clear.
- **During a burn:** the flame kept strong; layers added as the last coats with ash; nothing added that is painted, treated, or coated.
- **After a burn:** the char quenched until it floats; drained; taken to the store damp; the ash and the ground checked cold before the kiln is left.
- **Crushing:** wet, in closed bags, with respirator and goggles; then sieved; the fines kept for the seat, the coarse for the garden pile.
- **Seasonal:** feedstock gathered from the clearing and stacked to dry under cover a season ahead; the kiln checked for cracks and rust.
- **Annual:** the burn record — date, feedstock, moisture, char out — carried into char_yield; the feedstock the clearing gave, against the need.

How the programme relates to the kiln is described in the programme modules, in standing format.

8. Failure modes

From the design, not from experience.

Failure	Early sign	Repair and cost
Heavy smoke	smoke above the rim	smaller layers; drier wood
Wet feedstock	moisture reading	no burn; dry a season
Fire escapes the kiln	[from L2]	the strip, water, never alone
Char not fully charred	brown ends	back into the next burn
Dust when crushing	[from L2]	crush wet, in bags
Char heats in the store	[from L2]	damp, loose, spread
Clearing gives too little	burn record	garden char cut first
Burn refused or banned	written answer	the soil pit; bought char

9. What we got wrong

Nothing yet: this module has not been built.

10. Regulatory notes

Country	Instrument, number, date	Checked by, date
Ukraine	found 2026-09-29, not checked — see below	—
Romania	not checked	—
Poland	not checked	—
Slovakia	not checked	—
Hungary	not checked	—
Serbia	not checked	—
Czechia	not checked	—

Ukraine, found and not checked. Instruments found on 2026-09-29 and read on secondary copies and official summaries, because the statute site could not be reached; each is read on zakon.rada.gov.ua and signed before this row counts as checked.

- **Fire Safety Rules of Ukraine, Order No. 1417 of the Ministry of Internal Affairs, 30.12.2014.**
Point 1.20: a fire site at least 30 m from buildings, 50 m from coniferous and 25 m from deciduous forest, inside a cleared strip at least 2.5 m wide; open fire never left unattended.
- **Fire Safety Rules in the forests of Ukraine, Order No. 278 of 27.12.2004:** no fire in the forest in the fire-danger period; no burning of residues after ten or more days without rain.
- **Law No. 2707-XII of 16.10.1992,** on the protection of atmospheric air: Art. 20 bans burning waste in settlements except in special installations; Art. 11 asks an emissions permit of business entities for stationary sources. No act addresses a small kiln; commercial charcoal kilns hold permits, and some have been prosecuted without.
- **Code of Administrative Offences, Art. 77-1:** unauthorised burning of vegetation, 3,060 to 6,120 UAH for a citizen, doubled on land of the nature-reserve fund. A kiln fed with cut wood is arguably not burning off vegetation; that is untested.
- **Wartime.** Regional military administrations have ordered bans in forests at high fire danger. The state forestry agency says burning dry vegetation in wartime may be treated as sabotage; not verified in the Criminal Code. Local orders are asked for, not assumed.

- **Forest Code, No. 3852-XII of 21.01.1994**, Art. 69: any felling on forest land, sanitary felling included, needs a felling permit, issued free through the online environmental service. Whether an owner's collection of brushwood on forest land is free use is not verified.
- **Law No. 1436-III of 10.02.2000**, as amended in 2019: no clear felling in the Carpathian mountain fir-beech forests.
- **Cabinet Resolution No. 1045 of 01.08.2006**: trees on a homestead plot inside a settlement are removed at the owner's decision; the point number is not verified.

The written enquiries: to the regional environmental inspectorate and the regional emergency service, on whether the kiln is a burning of waste, a burning of vegetation, or a source needing a permit, and on the distances and fire-danger days; to the land register, on whether the wooded ground is forest-fund land; to the regional forestry authority, on brushwood; to the village council and the regional military administration, on local rules and orders. No burn is made before they are answered.

11. Adaptation notes

The design is a seed, not a copy: a different place grows a different form from it.

- **A drier valley**: more fire-danger days and fewer burn days; burn in the wet season, store a year's char.
- **A colder valley**: feedstock dries more slowly; stack it two seasons ahead.
- **A smaller group**: fewer burns; the same kiln, or a smaller cone of 2 mm steel.
- **A larger group or a larger garden**: more burns, not a larger kiln, until the burn record says otherwise.
- **No steel, or no welder**: the same cone dug in mineral soil.
- **No clearing to speak of**: char bought in, to the European Biochar Certificate, and the toilets' share first.

12. Provenance and licence

- **Written by**: Michel Garand. [set — others as they choose to be named]
- **From whose knowledge**: the sanitation working paper, EN v0.2; Hans-Peter Schmidt and Paul Taylor, Kon-Tiki, the Biochar Journal, 2014, for the flame-curtain kiln and its published design; Gerard Cornelissen and colleagues, PLOS ONE, 2016, and Science of the Total Environment, 2023, for yield, emissions, and moisture; South Umpqua Rural Community Partnership, 2018, Wilson and

colleagues, 2024, and the Permachar Kon-Tiki manual, 2023, for practice; the International Biochar Initiative, 2010, for crushing and safety; the European Biochar Certificate, version 10.5E, 2025, for storage and quality; Ukrainian retail and service listings of 2026-09-29 for prices.

- **Traditional knowledge:** none used. The literature records trees not taken for building — forked, with "candles", or windthrown — and potash made in the region for bleaching linen [for the elders]; none of it enters the design until the elders have confirmed, corrected, or withheld it.
 - **Text:** CC BY-SA 4.0
 - **Drawings and designs:** [hardware licence — open decision]; the kiln's design is its authors', published as open source
 - **Measurement data:** none yet
 - **Gate 1:** household decision, 2026-09-29, recorded by Michel Garand — v0.1 may be published as it stands, at L1, in English, Ukrainian, and German
 - **Wartime review:** 2026-09-29, Michel Garand — v0.1 read against compendium Section 2; nothing places the site; no supplier, market, or town near the site is named; the distances in Section 4 are the instruments' minimums, not measurements of the site; accepted
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13. What to check

- **The written answers** — on the kiln, the wood, the land's category, and local orders — before the first burn.
 - **The distances** — every row of the siting table, measured with a tape on the walk.
 - **The feedstock** — what the clearing yields in a year, stacked and measured, against 1.4 to 7.0 t of dry wood.
 - **char_yield** — char out against wood in, from the first burns.
 - **char_density** — a known volume of the site's char, weighed.
 - **The steel thickness** — confirmed against Schmidt and Taylor.
 - **The char and cover store** — re-estimated for a year's char and cover.
 - **The garden's char** — from the compost module's piles as measured.
 - **The quench water** — litres a burn, from the first burns.
 - **The responsible adult** — hours and pay.
 - **The Ukrainian row** — every instrument read and signed.
 - **Quotations** for the lines in Section 6, dated, to replace the listings and estimates.
 - **From L2, once built:** build times and out-of-order consequences in Section 5; the burn record in Section 7; early signs and repair costs in Section 8; Section 9.
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